

Haricharan Balasundaram

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EDUCATION

Indian Institute of Technology, Madras

CGPA: 9.62/10.00, Department Rank 1

B. Tech. (Hons.) in Engineering Physics + M. Tech. in Electrical Engineering; Minor in Computing

Nov 2021 - May 2026

PUBLICATIONS [↗](#)

Accepted:

1. *Estimating Error in Natural Distribution Estimation* [↗](#)
H. Balasundaram, A. Thangaraj. *Annual Allerton Conference on Communication, Control, and Computing* 2025.
2. *Distribution Estimation with Side Information* [↗](#)
H. Balasundaram, A. Thangaraj. *International Symposium on Information Theory* 2026.
3. *Generalized Capacity Planning for the Hospital-Residents Problem* [↗](#)
H. Balasundaram, G. Limaye, M. Nasre, A. Raja. *Theoretical Computer Science* 2026.
4. *Stability Notions for Hospital Residents with Sizes* [↗](#)
H. Balasundaram, Krishnashree J. B., G. Limaye, M. Nasre. *Foundations of Software Technology and Theoretical CS* 2025.

Under Review:

5. *Semantic Smoothing for Language Models via Distribution Estimation and Embeddings* [↗](#)
H. Balasundaram, S. S. Narashiman, P. Mathur, A. Thangaraj. *Submitted to Information Theory Workshop* 2026.
6. *Breaking the $O(\sqrt{T})$ Cumulative Constraint Violation Barrier while Achieving $O(\sqrt{T})$ Static Regret in Constrained OCO* [↗](#)
H. Balasundaram, Karthick Krishna Mahendran, Rahul Vaze. *Submitted to Performance* 2026.
7. *Convex Optimization with Nested Evolving Feasible Sets* [↗](#)
Karthick Krishna M., H. Balasundaram, R. Vaze. *Submitted to NeurIPS* 2026.
8. *Learning to Transmit Over Unknown Erasure Channels with Empirical Erasure Rate Feedback* [↗](#)
H. Balasundaram, K. Jagannathan. *Submitted to Information Theory Workshop* 2026.

AWARDS AND ACHIEVEMENTS

- Sole undergraduate recipient of *Teaching Assistant Recognition Award (TARA)* 2025 in the EE Dept for serving as Head TA.
- Recipient of *Ms. Latha and Sampath Srinath* prize for *Highest CGPA* in semesters 3 and 4 in the Engineering Physics program.

RESEARCH EXPERIENCE

Estimating Error in Natural Distribution Estimation [Master's Thesis]

[Prof. Andrew Thangaraj](#), Department of Electrical Engineering, IIT Madras

Feb 2025 - Jun 2026

- Studied unavoidable error of **distribution estimation** over a large alphabet, where *natural estimators* incur large error.
- Improved unavoidable error using two models of side information: known 'guess' distribution and ordering information
- Reduced perplexity of bigram models using semantic similarity between embeddings such as Word2Vec and GloVe.

Constrained Online Convex Optimization with Adversarial Constraints

[Prof. Rahul Vaze](#), School of Technology and Computer Science, Tata Institute of Fundamental Research

May 2025 - Present

- Working on **Constrained Online Convex Optimization**— simultaneous minimization of regret and constraint violation.
- Improved analysis for the existing algorithm, thus breaking the long-standing $O(\sqrt{T})$ —constraint violation bound in 2D.
- Proved that *Convex Optimization with Nested Evolving feasible Sets* is solvable under the assumption of strong convexity.

Learning to Transmit over Unknown Erasure Channels [Bachelor's Thesis]

[Prof. Krishna Jagannathan](#), Department of Electrical Engineering, IIT Madras

Dec 2023 - May 2025

- Developed algorithms for *transmission over erasure channels* with unknown erasure rates, using limited empirical erasure feedback.
- Analyzed two strategies: *Estimate-then-Transmit* (1 query, $O(T^{2/3})$ regret) and *Windowing* (logarithmic queries, $O(\sqrt{T})$ regret).
- Proved theoretical guarantees and validated performance through simulations, showing applicability to low-feedback systems.

Approximation Algorithms for Hospital-Resident Matchings

[Prof. Meghana Nasre](#), Department of Computer Science and Engineering, IIT Madras

Oct 2023 - Nov 2024

- Developed *approximation algorithms* and *inapproximability results* for quota augmentation in the **Hospital-Residents** setting.
- Analyzed *Hospital-Residents with Sizes (HRS)* problem, studying occupancy-stability and designing approximation algorithms.

Many-to-oneness of Lattice Filters

Prof. C. S. Ramalingam, Department of Electrical Engineering, IIT Madras

Sep 2023 - Nov 2023

- Utilized MATLAB to carry out brute-force calculations to determine the *oddness or evenness* of lattice coefficients.
- Explored conditions on one-oneness, and many-oneness of the mapping from lattice coefficients to transfer functions.

Control Systems for Rehabilitation

Prof. Sourav Rakshit, Gait and Motion Analysis (GAMA) Lab, Machine Design Section, IIT Madras

Sep 2022 - Jan 2023

- Applied LQR, and SAC control systems for trajectory tracking in **gait training** of paralyzed patients, achieving 75% accuracy.
- Created Python package [🔗](#) and contributed to *open-source repository* for implementing LQR to achieve position control.

TEACHING EXPERIENCE

- Head Teaching Assistant (TA)** for Signals and Systems (EE1101), coordinated 400+ students and 6 faculty– **TA award**.
- TA** for Multirate Digital Signal Processing, Probability Foundations, Statistics– supervised assignments and tutorial sessions.
- Conducted workshop on Cryptography in *Shaastra 2023* and conducted workshop on Probability and Finance in *Shaastra 2024*.
- Conducted information session on *Fundamentals of Mathematics and Programming* to incoming freshers in 2023.

RELEVANT COURSEWORK

EE:	Information Theory	Estimation Theory	Detection Theory	Convex Optimization
	Multirate DSP	RF and Optical Comms.	Communication Networks	Linear Algebra
	Probability and Statistics	Mathematical Physics	Advanced Topics in Communications (5G)	
Minor (CS):	Approximation Algorithms	Parameterized Algorithms	Graph Algorithms	Linear Programming

COURSE PROJECTS

EE5111: Estimation Theory [\[Slides\]](#)

Prof. Sheetal Kalyani, Department of Electrical Engineering, IIT Madras

May 2025

- 'Source Enumeration using Linear Shrinkage Coefficients'– addressed limitations of existing methods in *low SNR* regimes.
- Introduced a novel *running average-based filter* on the shrinkage coefficients to enhance detection accuracy.
- Achieved better results than existing methods for *source enumeration* under colored noise, with applications to 5G communications.

EE5121: Convex Optimization [\[Poster\]](#)

Prof. Uday Khankhoje, Department of Electrical Engineering, IIT Madras

Nov 2023

- Poster presentation on the paper '*Subsampled Hessian Newton methods for solving supervised learning problems*'.
- Achieved *12% improvement* on overqualified constraint datasets by integrating approximate Hessian direction with the gradient.

POSITIONS OF RESPONSIBILITY

Head and Founder

[Mathematics Club](#), Centre for Innovation, IITM

Nov 2022 - Mar 2024

- Co-founded and headed the *Mathematics Club* at IIT Madras, building a community with a reach of 1000+ students.
- Conducted sessions on number theory, probability, and linear algebra and headed projects on probability and group theory.

PROFESSIONAL EXPERIENCE

Software Developer Intern at D. E. Shaw India

Using LLMs to Automate Processing Vendor Emails

May 2024 - Jul 2024

- Built Python pipelines to process and validate large-scale financial data, improving speed and accuracy of business operations.
- Leveraged LLMs to automate vendor *email classification* and data extraction, reducing manual effort and streamlining workflows.

EXTRA-CURRICULAR ACTIVITIES

- Candidate Master* on Codeforces Competitive Programming; Global Rank #59 in Round #886 among 25,000 participants.
- Bronze Medal* in the Inter-IIT Meet Quant Competition (Dec 2023).
- Heading the *Communications* problem at Inter-IIT Tech Meet '25.
- Member of the Quiz Club IITM, winner of Shaastra SciTech Quiz '25.
- Strategist* in Programming and Cybersecurity clubs.
- Selected for ICPC Regionals at Amritapuri.
- Correspondent for *The Fifth Estate*, IITM's newsletter.